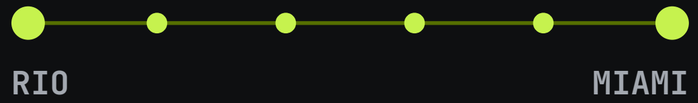
TESTNET · TEST KEYS

Radox Custody Rail

One record for the goods, from origin to delivery, that every company along the way will share.

Scan: the live shipment
rio-miami-rail.vercel.app

Toward a SWIFT-like standard on Stellar · Pilot corridor: Rio–Miami



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PMF · THE PROBLEM

Who holds the whole chain?

Warehouse — Port — Carrier — Customs — Recipient

The gap: each company records only its own leg, and no shared record follows the goods.

45M

ocean bills of lading issued a year

DCSA, Feb 2023

~11%

of bills of lading issued electronically, 2025

DCSA, as of 2025

22%

of recorded cargo thefts in 2025 involved insiders

TT Club & BSI Cargo Theft Report, Apr 2026

Ports run legacy systems: over 90% of ports in low- and middle-income countries have no port community system (World Bank remark relayed by IAPH, Nov 2023); IMO guidance plans for legacy systems with proprietary formats (FAL.5/Circ.54, 2024).

Context: our explainer video on a Rio–Miami trade-corridor initiative; this project is its custody layer.

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PMF · WHAT IT COSTS

Cargo goes missing. A bank got fooled.

The gap: a financier cannot see who holds the goods before paying.

R$900M

est. direct loss, 8,570 cargo robberies in Brazil, 2025

NTC&Logística survey, 28 Mar 2026

US$85M

HSBC's loss on forged documents for oil trades that never happened

Hin Leong case, Global Trade Review, 18 Nov 2024

US$2.5T

global trade finance gap (market size)

ADB survey of banks, Dec 2025

These figures size the market problem the rail addresses. Sources under each number.

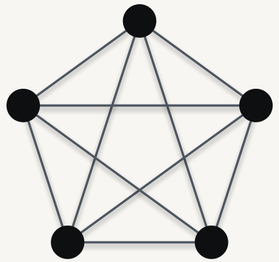
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THE PROPOSAL · A PUBLIC STANDARD

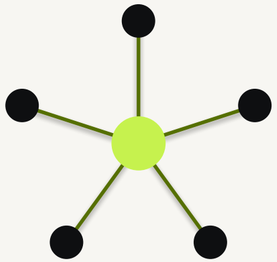
A SWIFT-like standard: integrate once

The goal is a shared, SWIFT-like custody standard that companies along a supply chain, from origin to destination, will use. **We are building it on Stellar, starting with the custody rail.**

PAIR BY PAIR · 10 LINKS



ONE STANDARD · 5 LINKS



Example, 5 companies: 10 custom links, or 5 links to one standard.

OPEN · ANYONE

— Record format, roles and events: the custody_registry contract (Apache-2.0)

— Verify any shipment, no login (slide 6)

RADOX WILL OPERATE · CLIENT PAYS

— Onboarding, issuance controls, operation of the rail

— The API and backend: closed, not self-hostable

— Per-institution subscription + fee per shipment (slide 9)

Each company integrates once instead of pair by pair. Radox is in conversations with ports, airports and chambers of commerce; legacy systems will join through one adapter to the Radox API.

Radox is the rail, never the issuer. Regulation and licences stay with the client and its local counsel.

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THE RAIL · LIFECYCLE BUILT

One rail, origin to delivery

One record will follow the goods across company boundaries; port and airport systems will connect by API, not a rewrite.



— **Rule of every link:** both sides sign one Stellar transaction (delivery: recipient and holder), or the call fails.
— Each step extends the on-chain `head`: anyone can replay the history from Horizon and check it against the head.
— The token stays with its holder in transit; only custody moves.

Built: this lifecycle on testnet, test keys. Next: real companies on the rail, the standard, integrations.

Keyed by sha256(container | origin); sits beside the carrier's bill of lading. [1] Radox policy, not enforced on-chain; redeem needs the holder's signature.

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THE PROOF · LIVE ON TESTNET

The lifecycle ran on testnet

Warehouse → Port, both signed	ledger 5381142 · 635c99…f0bd38
Port → Carrier, both signed	ledger 5381143 · 56a0f0…7b7cc2
Event attested by Source	ledger 5381144 · 333792…068905
Carrier → Customs, both signed	ledger 5381145 · ac268a…c3c4b2
Customs → Recipient, both signed	ledger 5381146 · 5199d4…400b7a
Delivered by Recipient to Holder, both signed	ledger 5381147 · 919def…144e37
Redeemed: token burned	864fd1…15644d

Verify head

`2b3d51f0` open · operator
`635c99ac` handover 1 · Warehouse → Port
`56a0f006` handover 2 · Port → Carrier
`ac268a49` handover 3 · Carrier → Customs
`5199d4fb` handover 4 · Customs → Recipient
`919def8c` deliver · Recipient → Holder
`864fd164` burn · existing redeem

7 successful

9 refused, e.g. wrong signer (simulated)

24 unit tests

MATCH head check, 25 Oct

`custody_registry`
`CCBJVXMJYWPHTDJHDFPRGKPGXGEP24JOPJAMUS5C3BMZJTRTWR2H4LII`

Also on testnet: handovers signed from separate computers (proofs/three-machine.json); two test companies' chains on one registry, sharing stops, each dual-signed (proofs/multi-chain.json).

Testnet, test keys.

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ECOSYSTEM · WHY STELLAR

Why Stellar: all signers, one call

01	**Multi-party auth in one call**	Soroban authorises both companies of a link inside one contract invocation: the handover is atomic.
02	**History from Horizon, checked by an on-chain head**	RPC keeps contract events about 7 days, so we replay each shipment's transactions from Horizon and check them against the head stored in the contract.
03	**Any Stellar asset**	The record stores a token address and never calls the token, so any tokenized Stellar asset can carry a custody record.

Stellar holds about US$4B in tokenized assets (SDF, as of 1 Sep 2026). **Nearest neighbours** (LumenLoop, SCF, Scout, 24–25 Oct): Kutana and Trustless Work release escrow on trade milestones; Pipeline finances commodity trade. None records a handover signed by two custodians; no SEP.

Open source, Apache-2.0: a custody primitive any Stellar tokenized-asset issuer can reuse. Radox will open API access to them, end to end.

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HACKATHON PROGRESS

Built this weekend, on disclosed prior work

PRIOR WORK · BEFORE SUN 25 OCT 09:00

— Radox platform (closed backend); its 3 Soroban contracts are on Stellar mainnet, public since 2026-08-19

— transfer_agent and its redeem (new testnet instance from the existing wasm)

— Architecture notes (20 Oct), planning files (24–25 Oct), the explainer video, our agent tooling (from 22 Oct)

— No contract code before then: first repo commit 25 Oct 09:00 UTC

BUILT AT HACKMERIDIAN

— custody_registry contract, 24 unit tests

— Lifecycle harness: 7 testnet transactions, 9 refused calls

— verify-head script, the public shipment page, this deck

— Radox API custody route reads the registry, read-only: same head (testnet)

AI attribution Code, tests, scripts, the shipment page, docs and this deck were made with AI coding agents (Claude Code); the team decided scope and what ships. Per-file record in AI_USAGE.md. The video narration is an AI voice generated from Pedro's voice.

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MARKET · MODEL · THE ASK

One rail, every corridor

MARKET · SOURCED FIGURES

US$94.2B

US–Brazil goods trade, 2025[1]

US$6.1B

Florida's goods exports to Brazil, its #1 market, 2025[2]

Global: US$26.26T world merchandise exports, 2025[3]; ~80% of goods trade by volume goes by sea[4]; ~45M ocean bills of lading a year[5].

MODEL

Subscription per connected institution + a fee per shipment.

Borrows the shape of Swift's billing rules: entry, annual, per-message usage[6]. Not affiliated with Swift.

Now running on testnet, test keys. **Next** the Rio–Miami pilot shipment, SCF Build, audit, mainnet, API for other Stellar assets.

[1] US Census Bureau, Trade in Goods with Brazil, 2025 (exports + imports, sum ours) [2] USTR, Florida trade data, 2025 [3] WTO, Mar 2026 [4] UNCTAD, Review of Maritime Transport (undated page, opened 25 Oct 2026) [5] DCSA, Feb 2023 [6] swift.com billing rules, accessed 25 Oct 2026

Our ask **Introductions to one financial institution and one port, airport or logistics operator for a first pilot shipment, and a Stellar engineer's review of the design.**

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THE TEAM

Why us: a founder already on Stellar mainnet

Pedro Wakigawa Saragossy

Founder, captain. Built Radox: 3 Soroban contracts on Stellar mainnet.

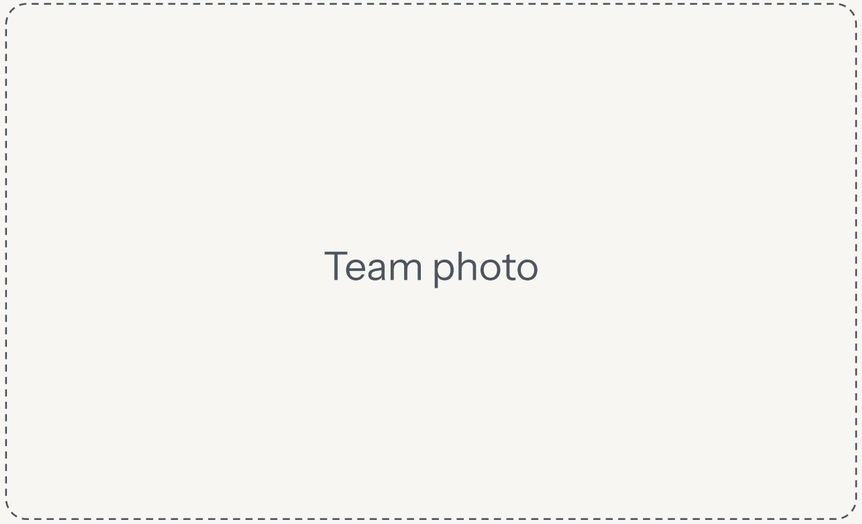
Leo Schlanger

Developer: custody_registry, the testnet harness and the proofs.

Vhendala

QA and product review: reads every deliverable as a judge would.

Leo and Vhendala joined for the hackathon.



Rio–Miami first. Our goal: one rail for goods, every corridor.

Live page rio-miami-rail.vercel.app **Code** github.com/pwsaragossy/rio-miami-rail (Apache-2.0)

AI Code, page, docs and deck made with AI coding agents (Claude Code). Narration: an AI voice from Pedro's voice. No AI-generated images in this deck.

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